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*Digital Differential Pressure Sensor Module, Calibrated, Temperature compensated,  $\pm 1500$  Pa, 16 bit ADC,  $\pm 3$  %FSS Pressure Accuracy.*

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*Datasheet - In Production*



## Description

The Siderion **SDP33 Airspeed** sensor is a compact, high-performance and factory-calibrated digital barometric pressure sensor (1PSI) ideal for precise altitude, pressure, and airspeed measurements.

It integrates a piezoresistive barometric sensor, a 16-bit ADC, and a DSP to cover a pressure range of  $\pm 1500$  Pa.

When used as an airspeed sensor in combination with a Pitot tube, it accurately detects differential pressure for airspeed calculations up to 180km/h at sea level in applications such as drones, aviation, and aerodynamics.

The sensor supports I<sup>2</sup>C communication interfaces, and does not need to be calibrated.

The SDP33 Airspeed sensor operates across a temperature range from -40°C to +70°C.

## Features

- 5V - 10mA power supply
- $\pm 1500$  Pa Pressure range
- Max 180 km/h at sea level
- I<sup>2</sup>C(up to 1MHz) interfaces

## Applications

- UAVs and Drones
- RC aircrafts
- Pneumatic controls
- Medical Instruments

Table 1. Device summary

| Order ref code | Temperature [°C] | Product size [mm] | Weight [g] |
|----------------|------------------|-------------------|------------|
| SID-SENS-SDP33 | -40 to +70       | 26.0 x 17.0 x 8.0 | 4          |

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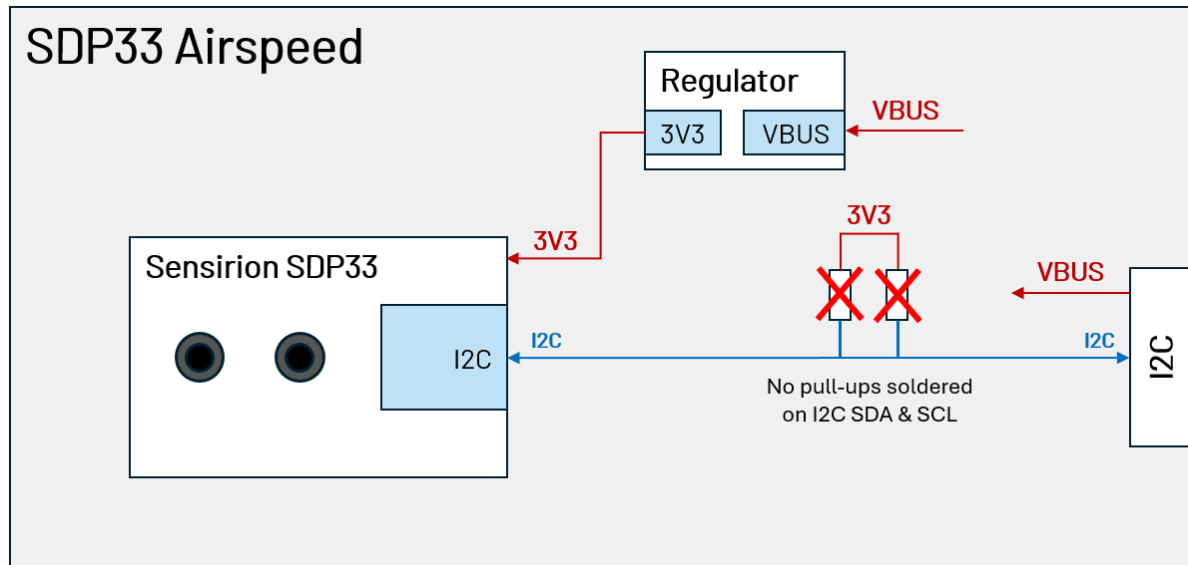
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# 1. Block diagram and pin description

## 1.1 Block diagram

Figure 1. SDP33 Airspeed sensor block diagram & connectivity (top view)



## 1.2 Pin description

Figure 2. SDP33 Airspeed sensor pinout (top & side views)



Table 2. Pinout configuration

|     | Pin | Name    | Type | Function         |
|-----|-----|---------|------|------------------|
| I2C | A1  | 5V IN   | P    | 5V input         |
|     | A2  | I2C SCL | I    | I2C Serial Clock |
|     | A3  | I2C SDA | I/O  | I2C Serial Data  |
|     | A4  | GND     | I    | Ground reference |

P : Power / I : Input / O : Output

## 2. Absolute maximum ratings

Stresses above those listed as “absolute maximum ratings” may cause permanent damage to the device. This is a stress rating only and functional operation of the device under these conditions is not implied. Exposure to maximum rating conditions for extended periods may affect device reliability.

Table 3. SDP33 Airspeed sensor absolute maximum ratings

| Symbol   | Parameter               | Maximum value   | Units |
|----------|-------------------------|-----------------|-------|
| Vcc      | Internal supply voltage | -0.5 to +5.5    | V     |
| Vdd      | I/O pins supply voltage | -0.5 to 3.3+0.1 | V     |
| Icc      | RF output current       | 10              | mA    |
| Sair_max | Maximum airspeed        | 180             | Km/h  |
| W        | Product weight          | 4               | g     |
| TOP      | Operating temperature   | -40 to +70      | °C    |
| TSTG     | Storage temperature     | -40 to +80      | °C    |



This device is sensitive to electrostatic discharge (ESD), improper handling can cause permanent damage to the part.

## 3. Specifications

### 3.1 Sensirion SDP33 specifications

Table 4. SDP33 mechanical and electrical specifications

| Symbol | Parameter                        | Test conditions                      | Min.       | Typ.        | Max.         | Units        |
|--------|----------------------------------|--------------------------------------|------------|-------------|--------------|--------------|
| Vcc    | 5V input                         |                                      |            | 5.0         |              | V            |
| Vdd    | Supply voltage for I/O           |                                      |            | 3.3         |              | V            |
| Idd    | Current consumption              | Measuring<br>Idle state              |            | 3.8<br>0.05 | 5.5<br>0.7   | mA<br>mA     |
| Idd_sm | Current consumption              | Sleep Mode (25°C)                    |            | 0.3         | 1            | µA           |
| Prange | Pressure range                   |                                      | -1500      |             | 1500         | Pa           |
| Pacc   | Pressure accuracy                | -500 to +500 Pa<br>-1500 to +1500 Pa |            |             | 3<br>6       | %FSS<br>%FSS |
| Pres   | Pressure resolution              | 16bit ADC                            |            |             | 0.045        | Pa           |
| Tacc   | Temperature accuracy             | -10°C to +60°C<br>-40°C to + 70°C    |            |             | ±2.0<br>±3.0 | °C           |
| Tres   | Temperature resolution           |                                      |            |             | 0.1          | °C           |
| Sair   | Measurable airspeed              | @ sea level                          |            |             | 50           | m/s          |
| Ctime  | Conversion time                  | Continuous Mode<br>Triggered Mode    | 0.45<br>40 | 0.5<br>45   | 0.55<br>50   | ms<br>ms     |
| Vil    | IO pin low level input voltage   |                                      | 0          |             | 0.8          | V            |
| Vih    | IO pin high level input voltage  |                                      | 2          |             | Vdd+0.3      | V            |
| Vol    | IO pin low level output voltage  | Iol = 0.6mA                          | 0          |             | 20%Vd<br>d   | V            |
| Voh    | IO pin high level output voltage | Ioh = 0.6mA                          | 80%V<br>dd |             | Vdd          | V            |
| Top    | Operating temperature            |                                      | -40        |             | +70          | °C           |
| I2Cadd | 7bit I²C address                 | 0b0100001 (0x21)                     |            |             |              |              |

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## 4. Applications

### 4.1 Pitot tube configuration

Figure 3. SDP33 Airspeed sensor

### 4.2 Airspeed computation (Bernouilli's equation)

For UAV and aeronautical use, the **SDP33** differential sensor is typically paired with a **pitot tube** to measure dynamic pressure ( $\Delta P$ ). This  $\Delta P$  is used to compute true airspeed (**TAS**) with the following relation:

$$v = \sqrt{\frac{2 \cdot \Delta P}{\rho}} \quad \text{Equation 1. Bernouilli's equation}$$

Where:

- **v** = airspeed (m/s)
- **$\Delta P$**  = differential pressure (Pa) measured by the sensor
- **$\rho$**  = air density (kg/m<sup>3</sup>), varies with altitude & temperature (**1.225 Kg/m<sup>3</sup>** at sea level)

With its fast response time and high resolution, the SDP33 enables accurate and real-time airspeed computation, even at low airspeeds ( $\pm 1500$  Pa range), critical for small UAVs and fixed-wing drones.

## 5. Communication interfaces

The sensor offers an I2C communication interface. All the inputs have internal pull-up resistors in normal operation and can be left open if not used.

All the PIOs are supplied by Vdd.

### 5.1 Slave I2C interface

An I2C compliant interface is available for communication with an external host CPU. The interface can be operated in slave mode only. It is fully compatible with Fast-Mode of the I2C industry standard. Since the maximum SCL clock frequency is 400 KHz (in Fast-Mode), the maximum bit rate is 400 Kbit/s. The interface stretches the clock when slowed down while serving interrupts, therefore the real bit rates may be slightly lower.

**The SDA and SCL lines do not have any internal pull-up resistors soldered in place. External pull-up resistors may be required for correct operation (typically 1 kΩ to 4.7 kΩ)**

Figure 4. I2C chronogram specifications

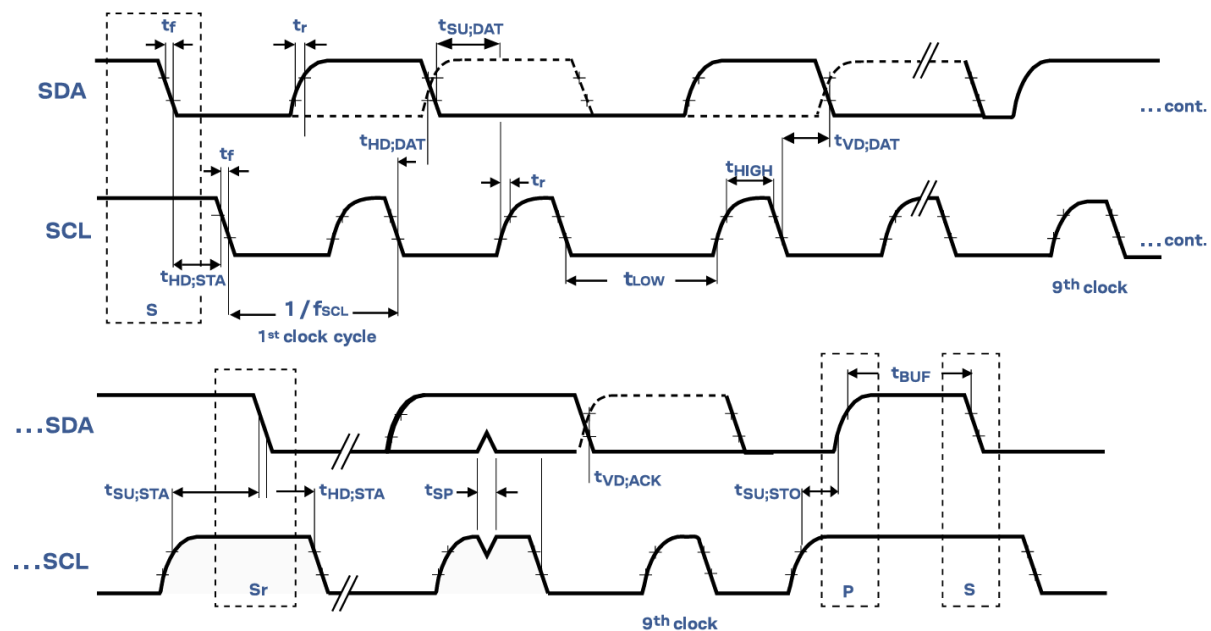
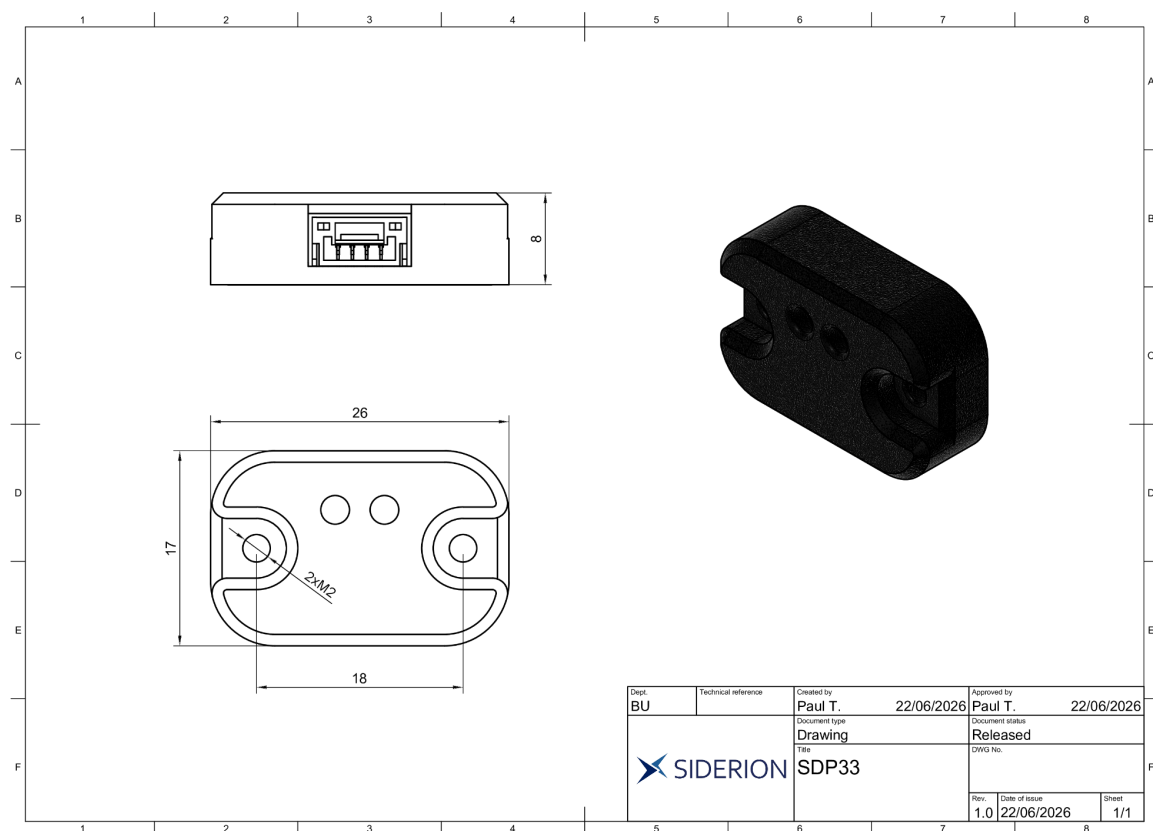


Table 5. SDP33 Airspeed sensor I2C timing specifications

| Symbol | Parameter                         | Min     | Max     | Units |
|--------|-----------------------------------|---------|---------|-------|
| Vil    | LOW-LEVEL input voltage           | 0       | 0.2xVdd | V     |
| Vih    | HIGH-LEVEL input voltage          | 0.8xVdd | Vdd     | V     |
| Cb     | Capacitive load for each bus line | 6       | 6       | pF    |
| Fscl   | SCL clock frequency               | 0       | 1       | MHz   |

## 6. Mechanical drawings

Figure 5. SDP33 Airspeed sensor v1.0 mechanical drawings



## 7. Revision history

Table 6. Document revision history

| Date         | Revision | Changes                             |
|--------------|----------|-------------------------------------|
| 23-June-2026 | 1.0      | SiderionDoc_SDP33 / Initial release |

## 8. Appendix

Sensirion SDP33 datasheet: [DP\\_DS\\_SDP33\\_D1.pdf](#)

Siderion user's guide:

Siderion tutorials:

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